

“THE GOAL OF **ENCYCLOPÉDIE** IS TO **ASSEMBLE** ALL THE **KNOWLEDGE... OF THE EARTH... SO THAT THE WORK OF CENTURIES PAST IS NOT USELESS.**”

Denis Diderot, French philosopher, in *Encyclopédie*, 1751



This illustration of a state-of-the-art scientific laboratory of the mid-18th century is from Denis Diderot's and Jean d'Alembert's *Encyclopédie*.

variations in depth as well as length and breadth. The positions or coordinates on these three axes, known as x, y, and z, are now central to trigonometry (see 1635–37). Also this year, Euler proved French mathematician Pierre de Fermat's theorem that certain **prime numbers**—numbers that are divisible only by themselves and the number one—can be expressed as the **sum of two square numbers**.

Meanwhile, French hydrographer **Pierre Bouguer** (1698–1758) was embroiled in a dispute concerning the shape of Earth. The French expedition to South America led by Bouguer and Charles de la Condamine in the 1730s (see 1733–39) had helped prove that **Earth's circumference is flattened at the poles**—but the pair disagreed bitterly on the exact results. Bouguer published his claim in *La figure de la terre* (*The Shape of the Earth*), in 1749. De la Condamine published his counterclaim two years later.

In 1750, English astronomer **Thomas Wright** (1711–86) began to think about the shape of the **Milky Way**, not then recognized as a galaxy. Wright speculated correctly that although we cannot see it because we are in the middle of it, the Milky Way is shaped like a flat disk.



Lightning charge
In Philadelphia in June 1752, experimenter and statesman Benjamin Franklin risked death as he proved lightning is electricity by flying a kite into a thundercloud to draw down the charge.

knowledge of the time, *Encyclopédie*. It was the first encyclopedia to include work from a variety of named contributors, and it aimed to collate the world's knowledge in one place.

In Edinburgh in 1751, Scottish physician

FRENCH MATHEMATICIAN PIERRE LOUIS MAUPERTUIS (1698–1759) wrote *Système de la Nature* (*The System of Nature*) in 1751. In it he discussed how characteristics are passed on from animals to their offspring, later the basis of the science of genetics. His ideas also foreshadowed naturalist Charles Darwin's once discredited ideas on pangenesis, an early theory of heredity now receiving renewed attention.

Also in this year, French philosophers **Denis Diderot** (1713–84) and **Jean d'Alembert** started work on their book that attempted to summarize all

Robert Whytt (1714–1766) discovered how the pupil of the eye **automatically opens or closes** in response to levels of light. His **pupil reflex** was the first discovery of a bodily reflex, an automatic response to a stimulus.

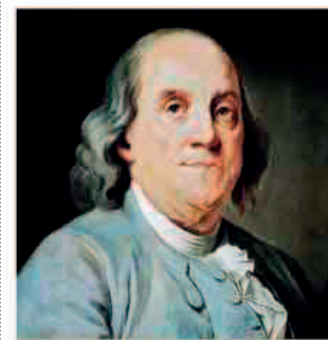
“HE SNATCHED THE **LIGHTNING** FROM THE SKY AND THE **SCEPTER** FROM **TYRANTS.**”

Anne-Robert Jacques Turgot, French economist and statesman, on Benjamin Franklin, in a letter to Samuel P. du Pont, 1778

Future American statesman **Benjamin Franklin** (1706–90) was intrigued by the similarity between lightning and the sparks in his home demonstrations of electrical phenomena. Franklin became convinced that **lightning is natural electricity** and in *Experiments and Observations in Electricity*, published in London in 1751, he described an experiment to prove his theory. This involved drawing lightning down to a spike on a sentry box.

In May 1752, Frenchman **Jean-Francois d'Alibard** (1703–99) tried Franklin's experiment in France and found that it worked. The following month, Franklin, not yet aware of d'Alibard's success, went out in a summer storm in Philadelphia to fly a **kite** under the clouds to **draw electrical charge** down the line to a key, insulated from the experimenter by a silk ribbon. As sparks streamed off the key, Franklin, like d'Alibard, could see that the **cloud was electrified**.

Also this year, physicist **Thomas Melvill** (1726–53) discovered that when he set different substances aflame, the flame gave **differently colored spectra** when shone through a prism. Salt gave a spectra dominated by bright yellow, for instance. This was the beginning of **spectroscopy**, by which substances are identified by the color of light they emit.



BENJAMIN FRANKLIN
(1706–90)

Born in Boston, Franklin lived most of his life in Philadelphia, where he ran a printing business. He was one of the founding fathers of American independence and became famous for his investigations into the nature of electricity. He also invented the lightning rod and a type of cast-iron stove, and he made studies of the Gulf Stream.

1750 Thomas Wright describes the Milky Way as shaped like a disk

1751 Robert Whytt discovers the pupil reflex

1751 Pierre Maupertuis suggests the theory of pangenesis

1751 Denis Diderot and Jean d'Alembert begin work on their *Encyclopédie*

1752 Benjamin Franklin proves lightning is electric

1752 English chemist Thomas Melville pioneers spectroscopy